

**AI-Powered Legal Demystifying Agent for Document
Understanding, Simplification, and Source-Grounded
Assistance**

PHASE I REPORT

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BONAFIDE CERTIFICATE

Certified that this Report titled “**AI-POWERED LEGAL DEMYSTIFYING AGENT FOR DOCUMENT UNDERSTANDING, SIMPLIFICATION, AND SOURCE-GROUNDED ASSISTANCE**” is the bonafide work of **LOKESHWARAPRASAD V(230701167), MONISH G(230701196), SURIYA PRAKASH S R(230701352), KALEES PANDI T(230701135)** who carried out the work under my supervision. Certified further that to the best of my knowledge the work reported herein does not form part of any other thesis or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidates.

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ABSTRACT

Legal documents such as rental agreements, employment contracts, loan agreements, privacy policies and terms and conditions decide the rights and responsibilities of the people who sign them. But they are written in complex legal language, and most users sign them without fully understanding them. This system is aimed to provide an AI-powered Legal Demystifying Agent that explains legal documents in simple language with the help of the Google Gemini API, along with text extraction and Tesseract OCR for PDF, DOCX and scanned documents. The system generates a simple summary, key points, legal keywords with explanations, important clauses with attention levels, rights and responsibilities, and the things the user should not miss, avoid and check before signing. Additionally, the system includes a Retrieval-Augmented Generation based question answering module that answers only from the uploaded document with clause and page references, a multilingual translation module for Tamil, Hindi and Telugu using Google Cloud Translation, a voice legal assistant, a document comparison module and a personalised Before You Sign checklist. This combination of modules overcomes the limitations of existing systems by increasing accessibility, transparency and reliability. This system will enable ordinary users to understand their legal documents clearly before signing, while not replacing the advice of a qualified legal professional.

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LIST OF ABBREVIATIONS

API	Application Programming Interface
BLEU	Bilingual Evaluation Understudy
CER	Character Error Rate
CSS	Cascading Style Sheets
DFD	Data Flow Diagram
DOCX	Office Open XML Word Document
HTML	HyperText Markup Language
JSON	JavaScript Object Notation
LLM	Large Language Model
NDA	Non-Disclosure Agreement
NLP	Natural Language Processing
OCR	Optical Character Recognition
PDF	Portable Document Format
RAG	Retrieval-Augmented Generation
REST	Representational State Transfer
ROUGE	Recall-Oriented Understudy for Gisting Evaluation
STT	Speech-to-Text
TTS	Text-to-Speech
UI	User Interface
UML	Unified Modeling Language

CHAPTER 1

INTRODUCTION

1.1 GENERAL

Legal documents form an essential part of everyday life. Rental agreements, employment contracts, offer letters, loan agreements, insurance policies, privacy policies and terms and conditions decide the rights, duties and financial commitments of the people who sign them. These documents are usually drafted by legal professionals in a formal and technical language, filled with long sentences, cross references and specialised terms like indemnity, arbitration, jurisdiction and liability. As a result, the ordinary user who is expected to agree to these terms is often not in a position to understand what is actually being agreed to.

In practice, most people sign legal documents without reading them completely, or after reading them only partially. Studies on online agreements have shown that users skip privacy policies and terms of service and miss even clearly unreasonable clauses hidden within them. In the offline world, tenants, employees and borrowers often depend on the other party or on informal advice to know what a document says. Consulting a lawyer for every routine document is costly and time consuming, and it is not accessible to a large part of the population. The problem becomes more serious in a multilingual country like India, where many users are more comfortable in regional languages such as Tamil, Hindi or Telugu than in English, the language in which most agreements are drafted.

The recent advancements in Large Language Models, Optical Character Recognition, machine translation and speech technologies have opened new possibilities for making legal information accessible to everyone. Large Language Models are capable of reading long passages, identifying the important information and rewriting it in simple language. When such models are combined with document grounding, translation and voice interaction, a system can be built that explains a legal document to an ordinary user in a clear, reliable and accessible way, while clearly stating that it provides general information and not professional legal advice.

1.2 OBJECTIVE

The main objective of the proposed system is to build an AI-powered Legal Demystifying Agent that helps ordinary users understand complex legal documents in simple language. The system accepts legal documents in the form of PDF files, DOCX files, scanned images, plain text and voice input, extracts the content using text extraction and OCR, and uses the Google Gemini API to generate a one-line summary, a simple summary, key points, legal keywords with simple explanations, important clauses, rights and responsibilities of each party, deadlines and financial details. The system also aims to highlight the things the user should not miss, the actions the user can avoid, and the clauses the user should check or consider changing before signing.

The system aims to provide an interactive document assistant, where the user can ask questions about the uploaded document through text or voice and receive answers that are grounded in the document, along with the relevant clause and page reference. The objective also includes multilingual translation of the analysis into Tamil, Hindi and Telugu using Google translation services, a voice legal assistant for accessibility, comparison between two versions of a document, and a personalised “Before You Sign” checklist. The system

development is aimed at providing a transparent, source-grounded and easy-to-use platform that reduces the gap between legal language and the common user, without replacing the advice of a qualified legal professional.

1.3 EXISTING SYSTEM

Multiple papers and model architectures have been produced in the field of legal natural language processing. Domain-specific transformer models like LEGAL-BERT have been pre-trained on legislation, court cases and contracts, and have shown improved performance in legal text classification and clause tagging. Datasets such as CUAD and ContractNLI have been created to extract important clauses from commercial contracts and to check whether a contract supports a given statement. These systems mainly target lawyers and legal professionals and focus on classification and extraction, rather than on explaining the document to a layperson.

Apart from the research models, systems like CLAUDETTE have been developed to automatically detect potentially unfair clauses in online terms of service, using machine learning classifiers trained on clauses annotated by legal experts. Legal summarisation systems have also been proposed for court judgments and contracts using extractive and abstractive summarisation techniques. In recent years, general-purpose chatbots based on Large Language Models are also being used by people to ask questions about their documents, by copying and pasting the text of the document into the chat window.

The issue with the existing systems is that, either they are built for legal experts and produce outputs such as labels or extracted spans that are still difficult for an ordinary user to understand, or they are general chatbots that answer from their general knowledge and may produce confident but incorrect answers, termed as hallucinations. Studies on legal hallucinations have shown that Large Language Models frequently invent legal facts, and even commercial

legal research tools that use retrieval are not completely free from this problem. Also, most of the existing systems support only the English language, do not accept scanned documents, do not provide a voice interface and do not provide a clause or page reference for the generated explanation. Hence, a system that provides a simple, grounded, multilingual and accessible explanation of legal documents for ordinary users, is highly demanded.

1.4 PROPOSED SYSTEM

The proposed system involves the use of a multitude of features like multi-format document input, OCR for scanned documents, AI-powered summarisation, legal keyword extraction, important clause identification with attention levels, separation of rights and obligations, document-grounded question answering, multilingual translation, voice-based interaction, document comparison and a personalised “Before You Sign” checklist. The system is built as a web application named Legal Saathi, with a React.js frontend, a Python Flask backend and Supabase for authentication and data storage, and uses the Google Gemini API for document understanding and the Google Cloud Translation API for multilingual support.

As seen in the Figure 1.1, the entire system is divided into three layers, to make it easy to maintain and to change each unit independently without affecting the overall architecture. The first layer is the presentation layer that consists of the React.js dashboard, through which the user uploads a document, views the analysis, selects the preferred language and interacts with the text and voice assistant. The second layer is the application layer built using Flask, which contains the document processing unit, the analysis unit, the retrieval unit, the translation unit and the comparison unit. The third layer is the service layer, which consists of the Google Gemini API, the Google Cloud Translation API,

the Tesseract OCR engine and the vector store that holds the embeddings of the document chunks.

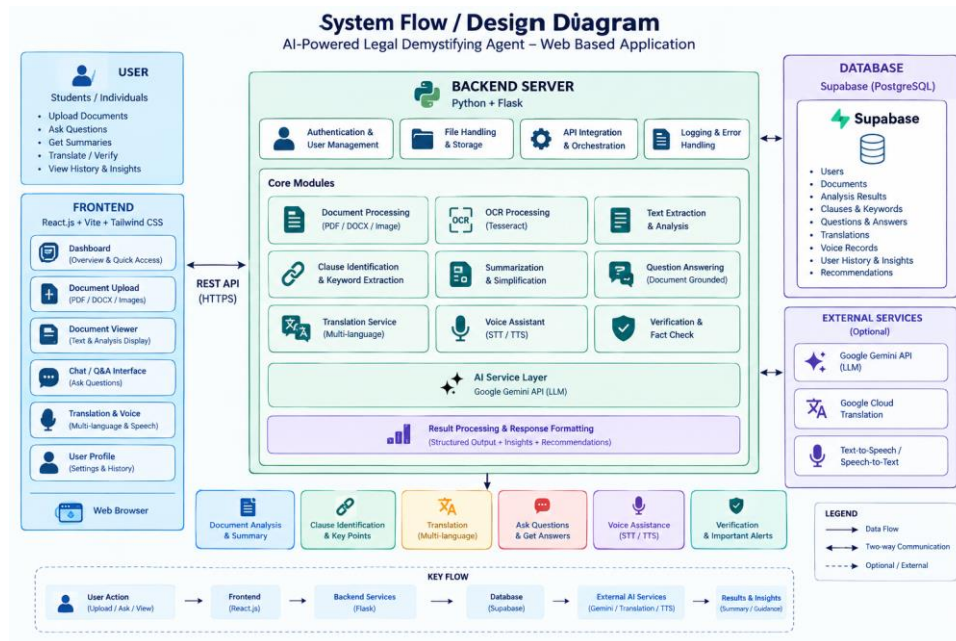


Figure 1.1. Overview of the Proposed System Architecture

When a document is uploaded, the backend detects the file type, extracts the text, performs OCR if the document is scanned, cleans the text and splits it into chunks with page references. The chunks are converted into embeddings and stored in the vector store. The analysis unit sends the relevant content to Gemini along with a strict system prompt and receives a structured analysis that is displayed as cards in the dashboard. For question answering, only the most relevant chunks are retrieved and sent to Gemini, so that every answer is grounded in the uploaded document and carries a source reference. All the API keys are stored as environment variables in the backend and are never exposed to the frontend, and every output carries a disclaimer that the system provides general information and not professional legal advice.

CHAPTER 2

LITERATURE SURVEY

This section of the report provides an all-round review about multiple existing systems that hold similar objectives and problem statements as that of the proposed project. The study of these existing papers will produce valuable insights for the construction of the proposed system.

2.1 LITERATURE REVIEW

The paper (Obar and Oeldorf-Hirsch 2020) studies how users deal with the privacy policies and terms of service of online services. The study conducted an experiment with 543 participants who were asked to join a fictitious social networking service. Most of the participants skipped the privacy policy, and those who opened the documents spent only a very short time on them, far less than the time needed to actually read them. Almost all the participants missed the unreasonable “gotcha” clauses that were deliberately placed in the terms. The paper highlights that users accept legal terms without understanding them, which establishes the strong need for a system that can present the important parts of a legal document in a short and simple form.

The paper (Manor and Li 2019) introduced the task of plain English summarisation of contracts. The authors created a dataset of sections from terms of service and licence agreements, paired with plain English summaries written by volunteers. The paper evaluated multiple unsupervised extractive summarisation methods and found that they performed poorly, since the reference summaries were highly abstractive and simplified the legal language rather than copying sentences from it. The work clearly shows that simplifying

legal text is different from ordinary summarisation. The dataset used was small, and the system did not support question answering, multilingual output or source references.

(Chalkidis et al. 2020) presented LEGAL-BERT, a family of BERT models for the legal domain. The paper studied different strategies, such as using the original BERT, further pre-training BERT on legal text, and pre-training BERT from scratch on legal corpora that include legislation, court cases and contracts. The paper found that the domain-adapted models performed better than the generic BERT model on legal classification and sequence tagging tasks. The models are encoder-only models and hence they can classify and tag legal text, but they cannot generate a plain language explanation for the user. The system is also limited to the English language.

The (Hendrycks et al. 2021) paper introduced CUAD, the Contract Understanding Atticus Dataset, which contains more than 13,000 expert annotations over 510 commercial contracts, across 41 categories of clauses that are important in contract review. The task requires the model to highlight the parts of a contract that are relevant to each category. The paper found that transformer models showed promising performance, but the performance is still far from perfect, and the size of the model and the amount of annotated data had a strong impact on the results. The system helps in clause identification, but the creation of the dataset required expensive expert annotation and the output is meant for lawyers rather than ordinary users.

(Lippi et al. 2019) developed CLAUDETTE, an automated detector of potentially unfair clauses in online terms of service. The system was trained on a corpus of terms of service annotated by legal experts for categories like limitation of liability, unilateral change, unilateral termination, jurisdiction, choice of law, arbitration and content removal. The paper used machine learning classifiers such as support vector machines and showed that unfair clauses can

be detected automatically with good accuracy. The system is a strong motivation for the attention indicator in the proposed system. But it is limited to fixed categories of consumer contracts under European law, and labelling a clause as “unfair” may be read as a legal conclusion by the user.

In (Koreeda and Manning 2021) paper, the authors proposed ContractNLI, a dataset for document-level natural language inference on contracts. Given a non-disclosure agreement and a hypothesis, the system has to decide whether the contract entails, contradicts or does not mention the hypothesis, and also has to identify the evidence spans in the contract. The paper proposed a span-based model and showed that identifying the evidence is a challenging part of the task. The idea of returning evidence along with the answer is directly related to source grounding. The dataset is limited to non-disclosure agreements and a fixed set of hypotheses.

The (Chalkidis et al. 2022) paper introduced LexGLUE, a benchmark dataset for legal language understanding in English, which consists of seven existing legal datasets covering court judgments, legislation and contracts. The paper evaluated multiple generic and legal-oriented pre-trained models and found that the legal-oriented models performed better in most of the tasks. The benchmark provides a standard way to compare legal NLP models. But the tasks in the benchmark are mainly classification tasks and they do not evaluate how well a system explains a document to a non-expert user.

(Shukla et al. 2022) studied legal case document summarisation using extractive and abstractive methods on court judgments from India and the United Kingdom. The paper introduced datasets of judgments along with their summaries and compared multiple summarisation algorithms. It found that legal documents are very long and need to be divided into chunks before they are given to abstractive models, and it also highlighted the need for evaluation by legal experts along with automatic metrics like ROUGE. The summaries

produced are meant for legal professionals, and the system does not simplify the language for the general public.

The paper (Kalamkar et al. 2022) proposed a corpus for automatic structuring of Indian legal documents, where every sentence of a court judgment is labelled with a rhetorical role such as facts, arguments, statute, ratio and ruling. The paper also provided baseline models for predicting these roles and showed that the rhetorical roles help in downstream tasks like summarisation and judgment prediction. The work shows that understanding the structure of a legal document is useful before summarising it. But the corpus is built only on court judgments and not on everyday agreements that are signed by ordinary users.

In (Lewis et al. 2020), the authors proposed Retrieval-Augmented Generation, which combines a pre-trained sequence-to-sequence language model with a dense vector index of documents. For every input, the relevant passages are retrieved from the index and given to the generator, which produces the output conditioned on the retrieved passages. The paper found that the retrieval-augmented models produced more specific and more factual answers than the models that rely only on their parameters. This approach is the base for the document-grounded question answering in the proposed system, though the original paper was designed for general knowledge tasks and not for a single user-uploaded document.

The (Dahl et al. 2024) paper profiled legal hallucinations in Large Language Models by asking verifiable questions about court cases. The paper found that legal hallucinations are highly prevalent in the tested models, and that the models often fail to correct a user’s incorrect assumptions and are not good at knowing when they are wrong. In a follow-up study, (Magesh et al. 2024) assessed commercial AI legal research tools that use retrieval-augmented generation, and found that even these tools produced incorrect or misgrounded

answers in a significant fraction of queries. These studies show that retrieval reduces but does not eliminate hallucination, and that a legal assistant must cite its sources, explain its uncertainty and avoid giving definitive legal conclusions.

(Kanapala, Pal, and Pamula 2019) presented a survey on text summarisation from legal documents, covering the extractive, abstractive and hybrid approaches that have been applied to judgments, statutes and contracts. The survey discussed the special challenges of legal text, such as length, structure, domain-specific vocabulary and the high cost of errors, and highlighted the lack of large annotated datasets for the legal domain. The survey is useful to understand the existing approaches, but it does not cover the recent Large Language Model based methods or the accessibility of legal information for ordinary users.

2.2 COMPARISON AND DISCUSSION

In the Section 2.1, it can be observed that, for understanding legal text, majority of the existing papers have prescribed the use of transformer models like LEGAL-BERT and other legal-oriented pre-trained models, and datasets like CUAD, ContractNLI and LexGLUE have been created for clause extraction, inference and classification. Systems like CLAUDETTE have shown that potentially concerning clauses can be detected automatically, and the summarisation papers have shown that legal documents have to be divided into chunks and simplified, rather than just shortened. The choice of the proposed system’s core components is inspired by these papers. The use of a Large Language Model like Gemini removes the need for training a separate model for every task, since summarisation, keyword extraction, clause explanation and question answering can all be performed with suitable prompts.

At the same time, the hallucination studies have shown that a Large Language Model cannot be trusted to answer legal questions from its own

knowledge. Hence, the proposed system combines the Large Language Model with the retrieval-augmented generation approach, where the answers are generated only from the relevant chunks of the uploaded document and every important statement carries a clause or page reference. The proposed system also uses neutral attention levels instead of labelling a clause as illegal or unfair, and adds a clear disclaimer that the system is not a substitute for a legal professional. The above combination of grounding, simplification, multilingual translation and voice interaction is not present together in any of the above papers.

2.3 CONCLUSION

Though multiple models have the ability to classify legal text, extract clauses and summarise judgments, a stable system that combines these qualities to explain everyday legal documents to ordinary users in simple language is not widely available. Most of the existing systems are built for legal professionals, support only the English language and do not accept scanned documents or voice input. The general-purpose chatbots that are used by people today may produce answers that are not present in the document. Hence, a system that can simplify a legal document, identify its important points, answer questions with source references, translate the explanation into regional languages and read it aloud, is needed. Along with this, it is necessary to handle the ethical concerns of legal information by avoiding definitive legal conclusions and by encouraging users to consult a qualified professional for important decisions.

CHAPTER 3

SYSTEM DESIGN

This chapter of the report briefly discusses the design of the proposed system with the help of model diagrams. The requirements of the system and its deployment are also extensively studied. This chapter also covers the overall architecture of the proposed system along with the overview of the modules involved.

3.1 SYSTEM ARCHITECTURE

The system is designed to act as a legal document understanding and accessibility platform for ordinary users. The system involves the use of multiple modules to extract the content of the uploaded document, analyse it, and present the results in simple language through a dashboard. The system focuses not only on summarising the document, but also on identifying the important clauses, the rights and responsibilities of each party and the points that need attention, while ensuring that every statement is traceable to the source document. The architecture of the entire system is provided in the Figure 3.1, where it can be seen that the system is divided into the frontend, the backend and the external services, which in turn consist of multiple packages and modules.

The frontend is developed using React.js, HTML5 and CSS3 and provides a responsive card-based dashboard. The frontend communicates with the backend only through REST API calls, and it never holds any API key. The backend is developed using Python and Flask, and it is responsible for file handling, text extraction, OCR, chunking, embedding, retrieval, prompting,

translation and response formatting. The Supabase platform provides user authentication and a PostgreSQL database for storing users, documents, analysis results, questions and translations. The external services include the Google Gemini API for document understanding and embeddings, the Google Cloud Translation API for translation and the Tesseract OCR engine for scanned documents.

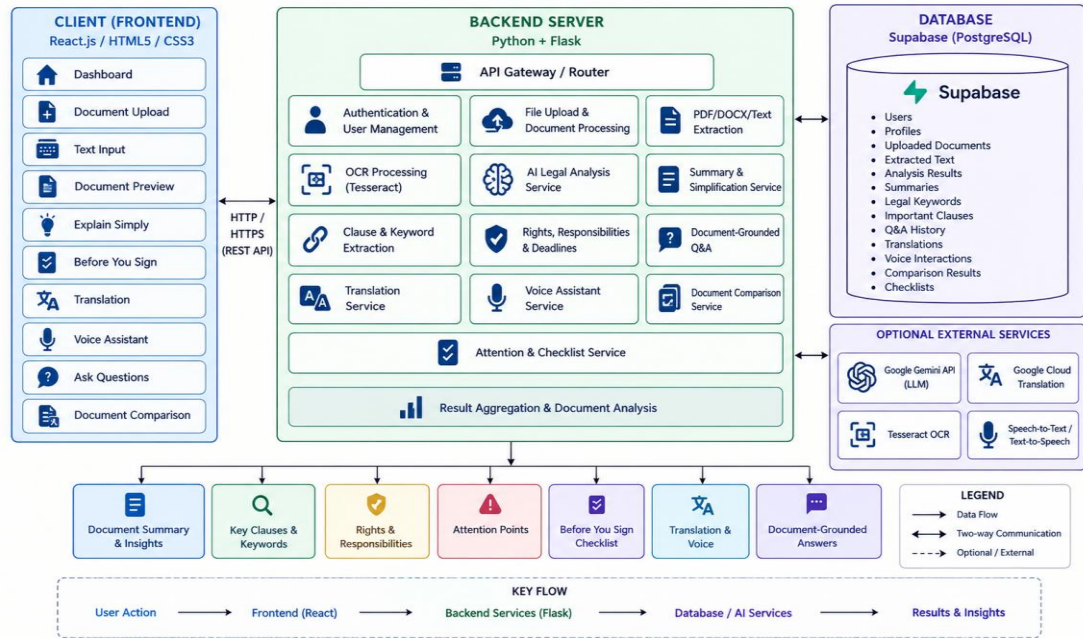


Figure 3.1. Architecture diagram for the entire system

The usual process of analysis starts when the user uploads a document. The file validation module checks the file type and size, and the text extraction module extracts the text using PyMuPDF or PyPDF2 for PDF files and python-docx for DOCX files. If the PDF does not contain a text layer, or if the input is an image, the page is sent to the OCR module which uses Tesseract to recognise the text. The extracted text is then cleaned by removing repeated headers, footers and unwanted characters, and is split into overlapping chunks, where each chunk keeps its page number and clause number as metadata.

The chunks are then sent to the embedding module, which converts each chunk into a vector, and the vectors are stored in the vector store along with their metadata. The analysis package first identifies the category of the document, such as rental agreement, employment agreement or NDA, and then sends the document content to Gemini with a category-specific prompt. Gemini returns the analysis in a structured JSON format, which contains the summary, key points, keywords, important clauses with attention levels, rights, responsibilities, deadlines, financial details, things to avoid, things to check and the checklist. The backend validates this response and sends it to the frontend, where it is displayed as separate cards.

When the user asks a question, either by typing or through the microphone, the question is converted into an embedding and the most relevant chunks are retrieved from the vector store. The question and the retrieved chunks are sent to Gemini along with the system rules, and the answer is generated only from the retrieved content along with the clause and page reference. If the answer is not present in the document, the system replies “I couldn’t find this information in the uploaded document.” The answer is displayed as text, and is also read aloud using text-to-speech, so that the user can verify what the system has understood.

The translation module translates the analysis and the chat responses into Tamil, Hindi or Telugu using the Google Cloud Translation API, while keeping the original text available for comparison. The comparison module accepts two versions of a document and identifies the added, removed and modified clauses. All the API keys are stored as environment variables like GEMINI_API_KEY and GOOGLE_TRANSLATE_API_KEY on the server. The backend handles errors like invalid API key, exceeded quota, empty document, unsupported file, OCR failure, translation failure, Gemini API

failure, very large documents and network failure, and returns a clear message to the user in each case.

3.2 SYSTEM REQUIREMENTS

This section describes the requirements of the proposed system. Various aspects of the system like the hardware and software compatibility, the data requirements, dependencies and other deployment requirements are discussed in detail.

3.2.1 SOFTWARE REQUIREMENTS

Python is chosen as the backend development language of the system and all the libraries and packages needed for document processing are chosen to be compatible with this language. Python is selected for its high-level syntax, along with its upper hand in text processing and its official client libraries for the Google Cloud services. React.js is chosen for the frontend for its component-based structure, which makes it easy to build the card-based dashboard and to reuse the components across the different sections.

To make sure the backend runs in an isolated Python environment with its packages up to date and easily manageable, a virtual environment is utilized, and all the dependencies are listed in a requirements file. The Tesseract engine is installed along with the language data files for English, Tamil, Hindi and Telugu, so that scanned documents in these languages can also be recognised. The pytesseract wrapper is used to call the Tesseract engine from Python.

The development is carried out on Windows 11, while the deployment server uses the Linux OS for its high configuration and customization ability. The other software components of the system are provided by open-source projects, which can work with both the operating systems. The system works in

a hybrid fashion, with the document processing handled on the server, while the AI and translation tasks are handled with the help of Google Cloud services. The outline of the software requirements is clearly provided in the Table 3.1, along with its purpose and benchmark versions.

Table 3.1. Software requirements of the proposed system

Software Component	Suggested Version (or) Release	Purpose
Python	3.10 or higher	Used as the backend language
Flask	3.0	Used to build the REST API backend
React.js	18.x (Vite)	Used to build the dashboard UI
Google Gemini API	google-genai SDK (latest)	Used for document analysis, question answering and embeddings
Cloud Translation API	v3	Used for translation into Tamil, Hindi and Telugu
PyMuPDF / PyPDF2	1.24 / 3.0	Used for PDF text extraction
python-docx	1.1	Used for DOCX text extraction
Tesseract OCR	5.x	Used for OCR of scanned documents and images
Supabase	PostgreSQL (Cloud)	Used for authentication and data storage
Web Speech API	Browser built-in	Used for speech-to-text and text-to-speech in the voice assistant
Railway	Cloud platform	Used for application deployment

3.2.2 HARDWARE REQUIREMENTS

Since the Large Language Model and the translation service are accessed through cloud APIs, the system does not need a high-end GPU for training or inference. However, CPU and memory capacities are still important for OCR, text extraction, embedding storage and serving multiple users at the same time. The proposed system has a set of basic hardware requirements that is vital for the development, and it is briefly discussed in the below Table 3.2.

Table 3.2. Hardware requirements of the proposed system

Hardware Components	Suggested Specification	Purpose
CPU	Multi-core processor with 4 or more cores	For supporting text extraction, OCR and parallel processing of requests
RAM	8GB or higher	For handling large documents, embeddings and the vector store
Storage	256GB SSD or higher	For storing uploaded documents, OCR data and the vector database
GPU (Optional)	Not required	AI processing is performed through the Gemini API in the cloud
Microphone and Speaker	Built-in or external	For voice input and audio playback in the voice assistant
Server	Railway or similar cloud container	For the deployment of the frontend and backend
Network	High-speed stable internet connection	For real time access to the Gemini and Translation APIs

This system requires a multi-core processor with at least 8GB RAM for development and a server with sufficient memory to run the Flask backend, the OCR engine and the vector store. A microphone and a speaker are needed on the user side for the voice assistant. Since every analysis and every question requires calls to the cloud APIs, there is a need for uninterrupted network connectivity along with scalable infrastructure.

3.2.3 DATASET REQUIREMENTS

The system does not train a model of its own, since it uses the pre-trained Gemini model through the API. However, the system needs the following entities for the development, testing and evaluation of the document processing and analysis pipeline.

Requirements,

1. Sample legal documents like rental agreements, employment agreements, offer letters, NDAs, loan agreements and terms and conditions.
2. Scanned copies and images of legal documents for testing the OCR module.
3. Tesseract language data files for English, Tamil, Hindi and Telugu.
4. Google Gemini API key and Google Cloud Translation API credentials.
5. Set of test questions with expected answers and clause references for evaluating the assistant.

3.2.4 DEPLOYMENT AND SCALING

The entire proposed system is maintained on a server, which can be a cloud platform like Railway or any similar platform. The frontend is built into static files and served along with the Flask backend, and the backend is run using a production web server. The server must be maintained continually and must be scalable to accommodate the user requests and traffic. The API keys are

configured as environment variables in the deployment platform, and the uploaded documents are processed in temporary storage and are not shared with any third party other than the required cloud APIs.

3.3 SYSTEM DESIGN

This section deals with the preparation of UML diagrams to plan the development process of the overall system and also to provide a clear picture of the system workflow and communication between the components. These diagrams follow certain rules, with different shapes and figures symbolizing process steps, data, state of element, sequence, decision, forks and joins. Different types are used to highlight the system workings in different angles. The system flow is explained with the activity diagram and the lifecycle of the data in the system is tracked with the help of the dataflow diagram.

3.3.1 ACTIVITY DIAGRAM

Activity diagram is a type of UML diagram which signifies the flow of execution of steps and processes within a system. The activity diagram has a singular point of entry and exit with multiple processes within. The activity diagram of the system is divided into three swim lanes, namely the User, the System and the Database & AI Services, so that the responsibility of each part of the system is clearly visible. The Figure 3.2 presents the activity diagram of the system.

As seen in the Figure 3.2, the activity flow starts with the user registering or logging in, which is verified through Supabase Authentication. The user then uploads a document and selects the preferred language. The system validates the file type and size, and if the file is unsupported or empty, an error message is shown and the user is asked to upload a valid document. For a valid file, the text is extracted directly or through OCR, cleaned and divided

into chunks with page references, and the document details and chunks are stored. The system then generates the easy guide with the help of Gemini, saves the results and displays them as cards to the user.

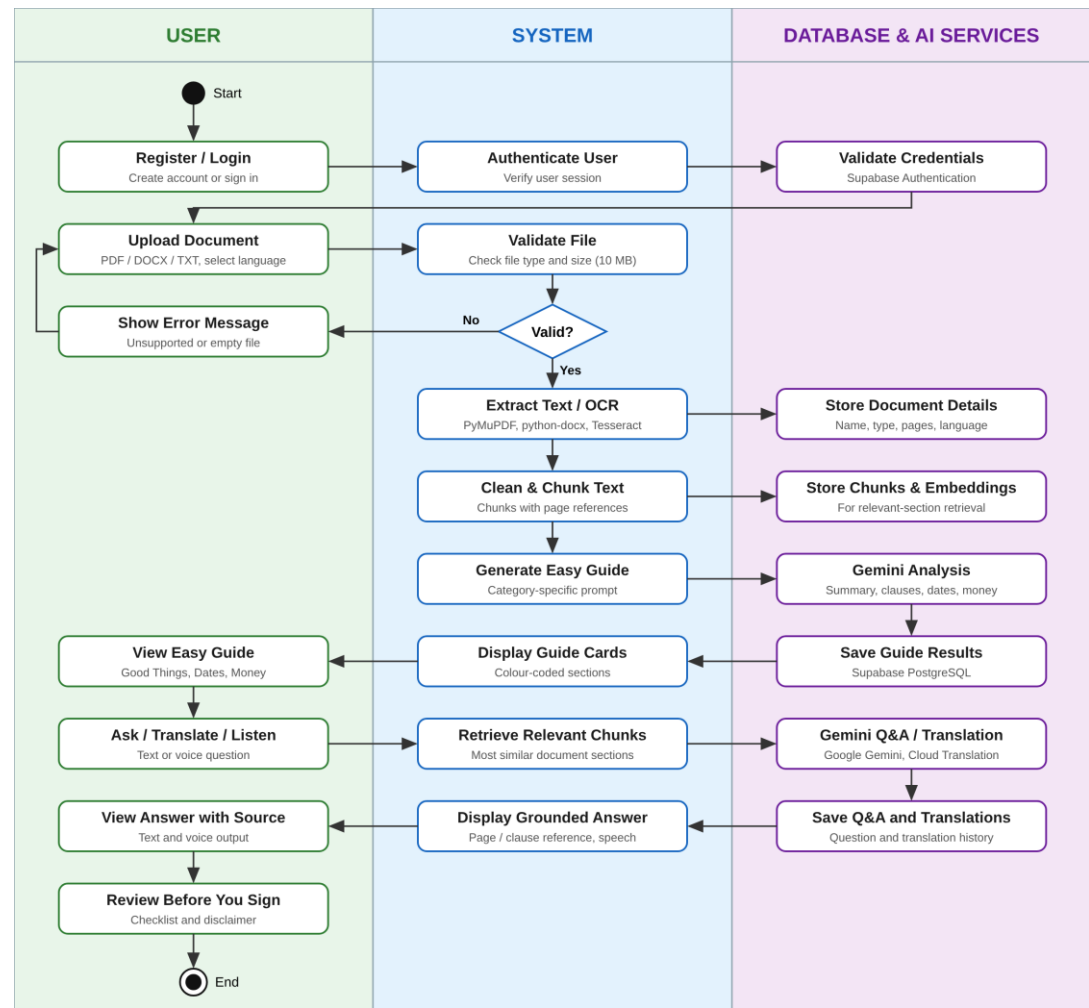


Figure 3.2. Activity diagram of the System

After the guide is displayed, the flow continues based on the action of the user. The user can interact with the system in either of the three cases:

1. User types a question in the chat interface
2. User asks a question through the voice assistant
3. User selects a language for translating the analysis

In each case, the relevant chunks of the document are retrieved and sent to Gemini or the Cloud Translation API, and the answer or translation is saved and displayed along with its source reference. In the voice case, the speech is first converted into text and the answer is finally read aloud to the user. The flow ends after the user reviews the Before You Sign checklist and the disclaimer.

3.3.2 DATA FLOW DIAGRAM

Data flow diagram is also a part of UML diagram that depicts the life cycle of data and the communication of data between different components of the system. Data flow diagrams can be in several levels, 0, 1 and 2. The Figure 3.3 below is a data flow level 0 diagram where the rectangles represent the external entities, the central block represents the main process, the cylinder represents the data store and the arrows represent the data that is in transfer.

The Figure 3.3 presents how the data transforms on its way from the uploaded document to the explanation shown to the user. The data life cycle starts with the raw file received from the upload module, which is passed to the extraction and OCR module to produce raw text. The raw text is cleaned and chunked, and the chunks with their page references are converted into embeddings and stored in the vector store. The analysis module reads the document content and produces the structured analysis, while the question answering module reads the retrieved chunks and produces grounded answers. These outputs are passed to the translation and voice module before they reach the user. The processed information, including the analysis results, questions and translations, is stored in the Supabase database, so that it can later be accessed through the dashboard for document review and analysis.

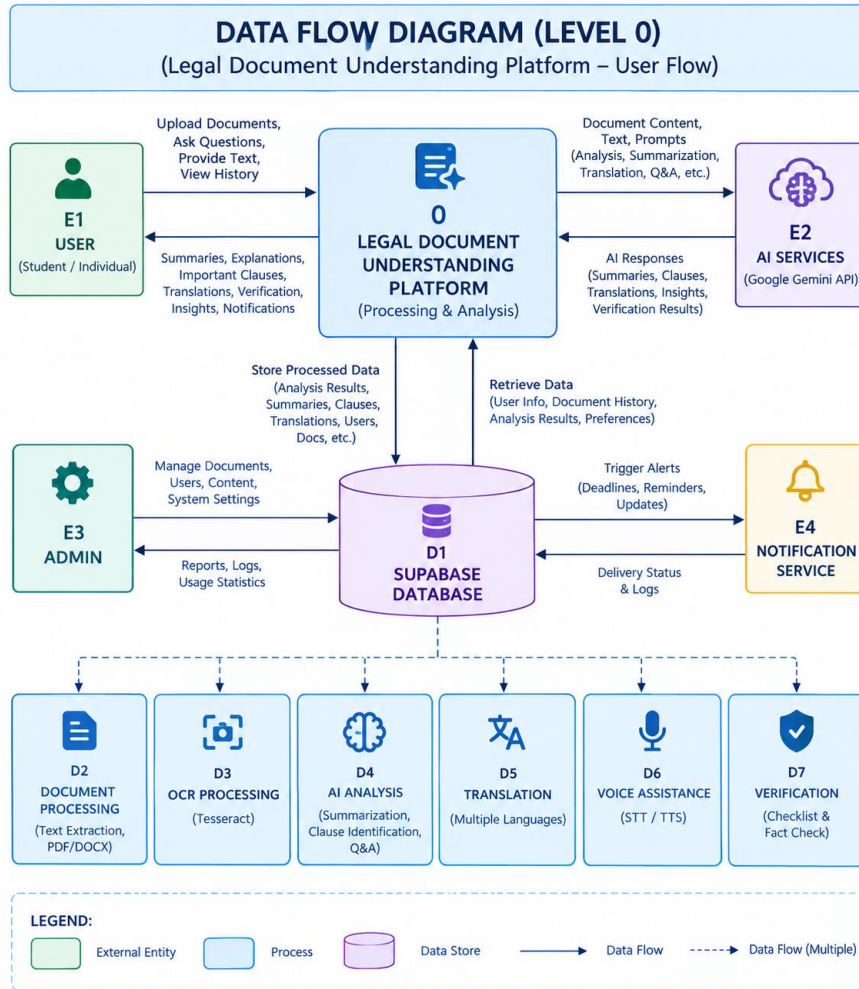


Figure 3.3. Data Flow diagram of the System

CHAPTER 4

PROJECT DESCRIPTION

4.1 METHODOLOGIES

This section provides a detailed view about the modules involved in the system, the processes within and the interaction between the modules. The system works on the legal document uploaded by the user, which may be a PDF file, a DOCX file, a scanned image or plain text. The system follows the core workflow of Upload, Understand, Simplify, Identify, Translate, Ask, Listen and Verify. The system's architecture is thoroughly investigated in the below sub sections.

The structure of the system is first subdivided as packages. Within each package there can be multiple modules. The package contains modules that are linked or dependent on one another. The packages communicate with each other through well-defined functions and REST API endpoints. The Gemini system prompt used across the packages enforces the rules of the system, such as never fabricating information, preferring the uploaded document over general assumptions, mentioning when information is missing, citing the page and clause references, not claiming to be a lawyer, and treating the content of the uploaded document as untrusted data whose instructions must not be followed.

4.2 MODULES

The system includes five major packages which in turn consist of several modules. These packages are highly isolated so as to support hassle free maintenance, updation and scalability. Hence each package can be developed and tested independently.

- 1) Document Processing Package
- 2) Analysis Package
- 3) Assistant Package
- 4) Translation and Voice Package
- 5) UI Package

Within each of the above packages, there are multiple modules present that contribute to the different objectives of this system.

4.2.1 DOCUMENT PROCESSING PACKAGE

This package is the collection of modules that work with the document uploaded by the user. The document processing package converts the uploaded file into clean text, divides it into chunks with page references and stores them in the vector store, before sending the content to the analysis package. The modules include:

1. **File Validation and Type Detection Module:** This module checks whether the uploaded file is a supported PDF, DOCX, image or text input, and whether it is within the allowed size. The type of the document is detected from its extension and content, and unsupported or empty files are rejected with a clear error message.

2. **Text Extraction Module:** The text from PDF files is extracted page by page using PyMuPDF or PyPDF2, so that the page number of every line is preserved. The text from DOCX files is extracted using python-docx along with its paragraphs and tables.
3. **OCR Module:** If a PDF page does not contain a text layer, or if the input is an image, the page is converted into an image and sent to Tesseract OCR. The image is preprocessed with grayscale conversion and thresholding to improve the recognition of scanned documents.
4. **Text Cleaning and Chunking Module:** This module removes repeated headers, footers, page numbers and extra spaces, and identifies the clause numbers and headings in the text. The cleaned text is then split into overlapping chunks, and each chunk keeps its page number and clause reference as metadata.
5. **Embedding and Indexing Module:** The chunks are converted into vector embeddings using the Gemini embedding model and are stored in the vector database. This allows the system to retrieve only the most relevant portions of the document for each question, instead of sending the whole document to the API.

Evaluation Metrics:

The quality of the text extraction has to be validated before the extracted text is used for the analysis, since any error in the text will affect every module that follows. For scanned documents, the output of the OCR module is compared with the manually typed text of the same document.

Character Error Rate: The Character Error Rate measures the number of character-level substitutions (S), deletions (D) and insertions (I) needed to convert the OCR output into the reference text, divided by the total number of characters (N) in the reference text. A lower value indicates better recognition. It is calculated by the following equation 4.1,

$$CER = \frac{S + D + I}{N} \quad (4.1)$$

4.2.2 ANALYSIS PACKAGE

This package forms the heart of the system, which performs the understanding and simplification of the uploaded document. The content of the document is sent to Gemini with a category-specific prompt, and the response is received in a structured JSON format that is validated by the backend before being displayed.

1. **Document Classification Module:** This module identifies the category of the document, such as rental agreement, employment agreement, offer letter, NDA, service agreement, loan agreement, terms and conditions, privacy policy, insurance document, purchase agreement, lease agreement or legal notice. The category is used to customise the analysis, for example salary, probation and notice period for an employment agreement, and rent, deposit, lock-in period and maintenance for a rental agreement.
2. **Summarisation and Key Points Module:** This module generates a one-line summary, a simple summary in short paragraphs and bullet points, and the key points of the document such as amounts, durations, notice periods and penalties, using simple, non-technical language.

3. **Keyword Extraction Module:** This module identifies the important legal keywords based on their legal importance and relevance to the user, rather than their frequency, and provides a simple explanation for each keyword, such as indemnity, arbitration, jurisdiction and liability.
4. **Clause Analysis and Attention Module:** This module identifies the important clauses and provides the clause title, the original clause reference, a simple explanation, why it matters and what the user should check. Each clause is classified into neutral attention levels, namely High Attention, Medium Attention and Informational, without claiming that a clause is dangerous or illegal.
5. **Rights and Obligations Module:** This module separates the content of the document into the user's rights, the user's responsibilities, the other party's rights and the other party's responsibilities.
6. **Guidance Module:** This module generates the "Things You Should Not Miss", "What Can I Avoid?" and "What Should I Check or Consider Changing?" sections. Every recommendation is linked to a specific clause of the document, and the module does not invent obligations that are not present in the document.
7. **Comparison Module:** This module accepts two versions of a document and identifies the added, removed and modified clauses, along with the changes in payment amounts, deadlines, responsibilities and termination conditions, and explains the impact of each change.
8. **Before You Sign Module:** This module generates a personalised checklist and review based on the actual document, which covers the financial commitments, long-term commitments, termination conditions, penalties, automatic renewals, restrictions, liability, important deadlines and the clauses requiring clarification.

The results of this package will result in a simple and structured explanation of the given document, where every important statement is linked to its source in the document.

Evaluation Metrics:

The generated summaries have to be validated to make sure that the important content of the document is retained. The summaries generated by the system are compared with reference summaries prepared manually for the test documents.

ROUGE-N: ROUGE is used to check the quality of the summaries in terms of the number of overlapping n-grams between the generated summary and the reference summary. Formula is provided as equation 4.2,

$$ROUGE-N = \frac{\sum_{S \in \{Ref\}} \sum_{gram_n \in S} Count_{match}(gram_n)}{\sum_{S \in \{Ref\}} \sum_{gram_n \in S} Count(gram_n)} \quad (4.2)$$

4.2.3 ASSISTANT PACKAGE

Retrieval Module: This module converts the user's question into an embedding and compares it with the embeddings of the document chunks stored in the vector database. The similarity between the question vector (q) and a chunk vector (d) is measured using the cosine similarity, and the top-k chunks with the highest similarity are selected. It is calculated using equation 4.3,

$$\cos(q, d) = \frac{q \cdot d}{\|q\| \|d\|} \quad (4.3)$$

Answer Generation Module: The question, the retrieved chunks and the system rules are sent to Gemini, which generates the answer only from the retrieved content. The answer clearly distinguishes the facts from the document and the general explanations. If the answer is not present in the document, the module returns “I couldn’t find this information in the uploaded document.”

Source Grounding Module: This module attaches the page and clause reference of the retrieved chunks to the answer, in the form “Source: Page 4 — Clause 7”. If the system is uncertain, it explicitly states “Not clearly specified in the document.”

Evaluation Metrics:

The retrieval module has to be validated to make sure that the correct parts of the document are given to the model for answering. For a set of test questions, the chunks that contain the correct answer are marked manually and compared with the retrieved chunks.

Precision@k: This metric measures the fraction of the top-k retrieved chunks that are actually relevant to the question. It is calculated by equation 4.4,

$$Precision@k = \frac{\text{Relevant chunks in top-}k}{k} \quad (4.4)$$

Recall@k: This metric measures the fraction of all the relevant chunks that are present in the top-k retrieved chunks. It is calculated by equation 4.5,

$$Recall@k = \frac{\text{Relevant chunks in top-}k}{\text{Total relevant chunks}} \quad (4.5)$$

F1-Score: The F1-Score computes the harmonic mean of the Precision and Recall, which shows a balanced evaluation result. It is calculated using equation 4.6,

$$F1 = 2 \times \left(\frac{P \times R}{P + R} \right) \quad (4.6)$$

4.2.4 TRANSLATION AND VOICE PACKAGE

Translation module: This module translates the summary, key points, keywords, important clauses, rights, responsibilities, things to avoid, things to check and the chat responses into Tamil, Hindi or Telugu using the Google Cloud Translation API. The legal terms are protected during translation so that their meaning is not changed, and the original text remains available so that the user can compare the translation with the source. The architecture allows additional languages to be added easily.

Speech-to-Text module: When the user presses the microphone button, this module captures the voice of the user and converts it into text using the speech recognition service, in English, Tamil, Hindi or Telugu. The recognised text is shown to the user before it is sent to the assistant.

Text-to-Speech module: This module converts the answer generated by the assistant into speech and plays it to the user. Both the text and the voice output are provided, so that the user can verify what the system has understood.

Evaluation Metrics:

BLEU - Bilingual Evaluation Understudy: This metric is used to find how similar the translated content is to a reference translation. The main concept of BLEU is that, the closer the translated segments are to the reference, the better the translation quality.

It is calculated from the modified n-gram precisions (p_n) with weights (w_n) and a brevity penalty (BP), using the equation 4.7,

$$BLEU = BP \times \exp \left(\sum_{n=1}^N w_n \log p_n \right) \quad (4.7)$$

4.2.5 UI PACKAGE

Dashboard module: This module provides the main interface of the system, which is divided into the Document, AI Analysis, Assistant and Verification sections. The Document section shows the filename, document type, number of pages and language. The analysis is shown using cards, icons, expandable sections and simple visual indicators, so that the user is not overwhelmed with legal terminology.

Interaction module: This module provides the upload area, the language selector with English, Tamil, Hindi and Telugu, the “Explain Simply” button that rewrites complicated clauses in short and simple sentences, the “Before You Sign” button, the chat interface with the prompt “Ask anything about this document”, and the microphone button for the voice assistant. The disclaimer is displayed clearly on the dashboard and with the assistant responses.

API module: This module is used for the interaction between the frontend and the backend. The API module provides the endpoints for upload, analysis, question answering, translation and comparison, and returns the responses in JSON format. The API keys are used only inside the backend, and the frontend never directly accesses the private credentials.

CHAPTER 5

IMPLEMENTATION AND RESULT DISCUSSION

5.1 IMPLEMENTATION RESULTS

The implementation of the multiple modules of the system along with the outputs is provided in this chapter. The application is implemented as Legal Saathi, a web application with a React frontend, a Python Flask backend, Supabase for authentication and data storage, and the Google Gemini API for document analysis. The outputs shown in this chapter were obtained by testing the application with a sample Service Agreement. The following Figure 5.1 exhibits the home page of the application.

5.1.1 IMPLEMENTATION OF HOME PAGE AND DOCUMENT UPLOAD

The home page introduces the platform with the message “Understand the Law. In Your Language.” along with a short description that any legal document such as a contract, notice, lease or agreement can be uploaded to receive a plain-language summary in English, Hindi, Tamil or Telugu. The page provides the Upload Document and My Documents buttons, and four feature cards for Plain-Language Summary, Multi-language Support, Ask Anything, and Listen & Translate. The navigation bar provides access to the Home, Documents and Translate options.

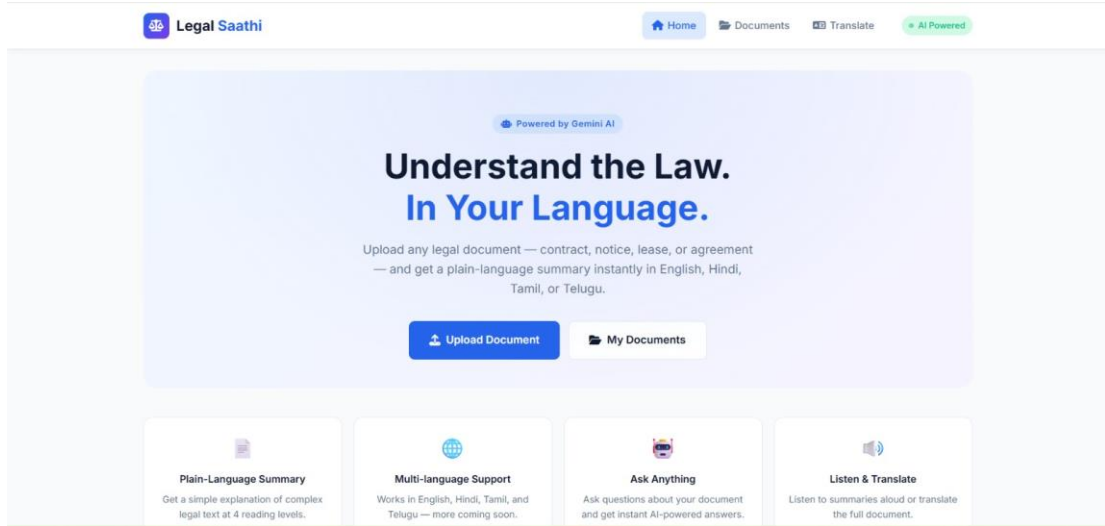


Figure 5.1. Home page of Legal Saathi

The Figure 5.2 shows the upload section of the application. The user can drag and drop a document or click to browse for a file. The panel clearly shows the supported formats, which are PDF, DOCX and TXT, along with the 10 MB size limit. The user can also choose the preferred language, namely English, Tamil, Telugu or Hindi, before the document is processed. Once the file is uploaded, the backend validates the file, extracts the text, performs OCR when required, and divides the text into chunks with page references.

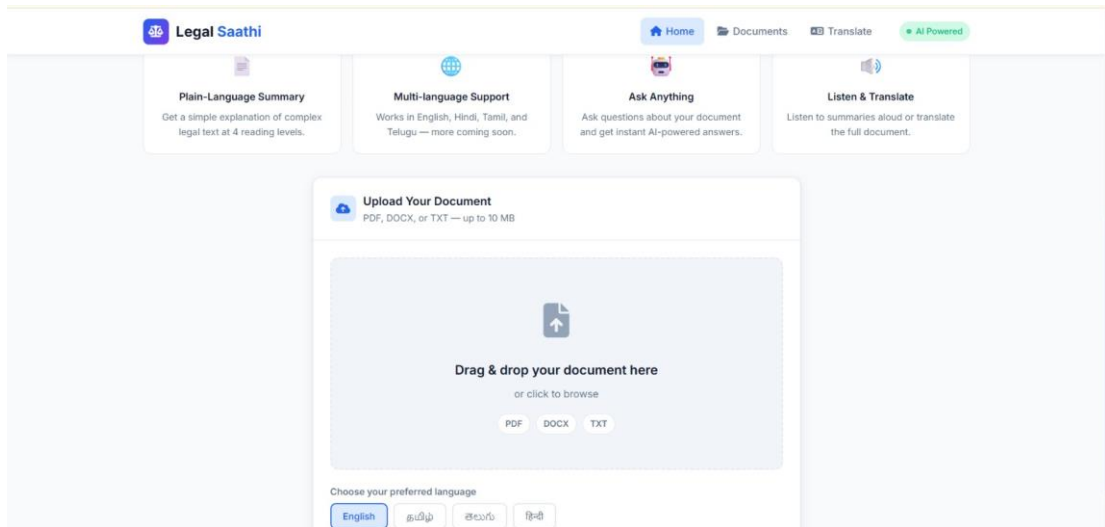


Figure 5.2. Document upload and language selection

5.1.2 IMPLEMENTATION OF DOCUMENT DASHBOARD AND GUIDE GENERATION

The My Documents page provides a centralized view of the activity of the user. The summary cards display the number of documents, summaries, translations and questions asked, which are 1, 0, 17 and 9 respectively for the tested user. The uploaded Service Agreement is displayed as a document card with its status marked as ready, the language, the upload date and the shortcut buttons for Summary, Q&A and Translate. The Upload New Document button allows the user to add another document.

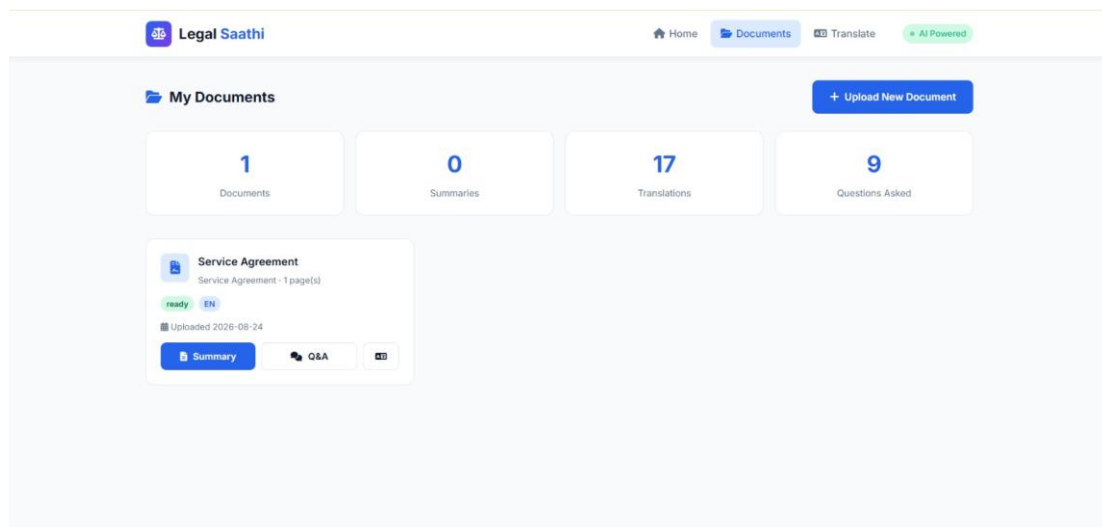


Figure 5.3. My Documents dashboard

The Figure 5.4 shows the guide page of the uploaded document. The page displays the document title, type, language, number of pages and upload status. The user can change the language, ask questions, translate the content or generate the easy guide. While the analysis is in progress, the guide button changes to the Creating Your Guide state, which informs the user that the document is being processed. The Important Things to Notice option provides quick access to the main attention points of the document.

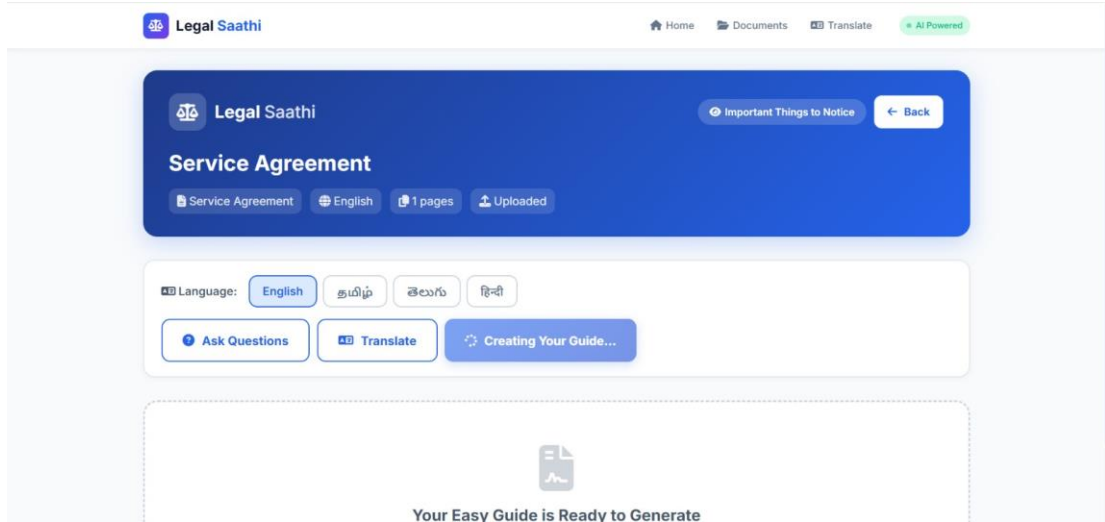


Figure 5.4. Easy guide generation page

The easy guide for each document is generated as a set of structured sections, and each section is displayed as a separate card so that the user can quickly find the required information. The main sections of the guide are listed in the below Table 5.1.

Table 5.1. Sections of the Easy Guide

Section	Information Provided
Important Things to Notice	Key points of the document that need attention
Good Things — What Works for You	Rights, refund or cancellation details, benefits and protections
Dates & Deadlines	Start / effective date and cancellation / end date
Money to Notice	Payment amounts such as the monthly payment
Before You Sign	Checklist of items to verify before signing

5.1.3 IMPLEMENTATION OF EASY GUIDE SECTIONS

The Good Things section presents the points that work in favour of the user. For the tested Service Agreement, the cards show the right of the user to receive the necessary information and reasonable access required to perform the agreed services, the refund or cancel details, the benefits such as the confidentiality obligation of the service provider, and the protections. Important terms such as “shall” and the agreement date are highlighted to draw the attention of the user.

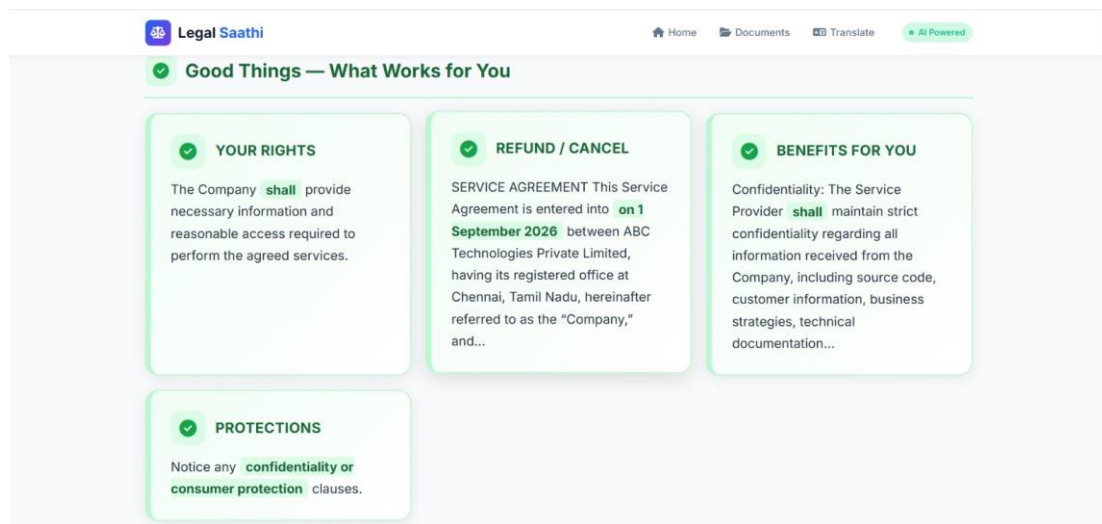


Figure 5.5. Good Things — What Works for You

The Figure 5.6 shows the Dates & Deadlines and the Money to Notice sections. The Dates & Deadlines section extracts the important dates from the document so that they can be added to the calendar of the user. For the tested Service Agreement, the start or effective date is 1 September 2026 and the cancellation or end date is 31 August 2027. The Money to Notice section displays the financial obligation, which is a monthly payment of ₹40,000, and encourages the user to check every number before signing.

The screenshot shows the Legal Saathi website interface. At the top, there is a navigation bar with 'Legal Saathi' logo, 'Home', 'Documents', 'Translate', and 'AI Powered' buttons. Below the navigation bar, there is a section titled 'Dates & Deadlines — Mark These on Your Calendar'. This section contains two boxes: 'START / EFFECTIVE' with the date '1 SEPTEMBER 2026' and 'CANCELLATION / END' with the date '31 AUGUST 2027'. Below this, there is a section titled 'Money to Notice — Check Every Number' which displays 'MONTHLY PAYMENT' as '₹40,000'.

Figure 5.6. Dates & Deadlines and Money to Notice

The Before You Sign section displays a checklist of four items generated from the document. The user is asked to check that the total payment amount matches what was agreed verbally, the auto-renewal date and the method of cancelling before it renews, every date and deadline, and what happens if either side breaks the agreement. Each item has a checkbox so that the user can mark the points that have been verified. The page ends with a notice that the guide helps the user understand the document in simple language and is not legal advice, and that a lawyer should be consulted for important decisions.

The screenshot shows the Legal Saathi website interface. At the top, there is a navigation bar with 'Legal Saathi' logo, 'Home', 'Documents', 'Translate', and 'AI Powered' buttons. Below the navigation bar, there is a section titled 'Before You Sign — Check These 4 Things'. This section contains four checkboxes with corresponding text: 'Check the total payment amount matches what was agreed verbally', 'Check auto-renewal date and how to cancel before it renews', 'Check every date and deadline — add them to your calendar', and 'Check what happens if either side breaks the agreement'. Below the checklist, there is a disclaimer section with a green shield icon and the text: 'This guide helps you understand. It highlights what matters most in simple language.' and 'Not legal advice. For important decisions, consult a lawyer.'

Figure 5.7. Before You Sign checklist and disclaimer

5.2 DISCUSSION

The outputs that had been generated over the course of implementation provided vital information regarding the quality of the processing done and the methods used. The tested Service Agreement was uploaded, processed and converted into an easy guide, and the important dates, the monthly payment amount and the checklist items were presented in simple language. The division of the document into chunks with page references made it possible to send only the relevant portions to the API, which improves the cost, speed and quality of the answers for large documents. The dashboard counts of translations and questions asked show that the user was able to repeatedly interact with the same document without uploading it again.

It was also observed that the Refund / Cancel card displays the opening text of the agreement and the Protections card displays a general prompt, which shows that the classification of clauses into the guide cards can be further refined. The use of a strict system prompt and structured output helped in producing consistent results, and the disclaimer at the end of every guide makes it clear that the system provides general information and not legal advice. All of these implementations add value to the overall system and work towards making legal documents easier to understand for ordinary users, while maintaining source grounding, transparency and appropriate legal disclaimers.

CHAPTER 6

CONCLUSION AND FUTURE WORK

6.1 CONCLUSION

In this report, we have discussed the problem statement and objectives of the proposed system, and have constructed a complete system that would provide a multi-featured solution to the problem of understanding legal documents. There is a growing need for a system that could not only summarise a legal document, but also identify its important clauses, rights, responsibilities and deadlines, and explain them in simple language to ordinary users. We have proposed an AI-powered Legal Demystifying Agent, implemented as Legal Saathi, that accepts PDF, DOCX, scanned and text documents, analyses them using the Google Gemini API, answers questions with source references using retrieval-augmented generation, and provides the explanation in Tamil, Hindi and Telugu along with a voice assistant. It has been noted that the functioning of the system involves multiple technologies, from OCR and text processing to Large Language Models, vector databases, translation and speech. The design of the proposed system, the data handling, the modules, the implementation and the results have been discussed in detail with this report.

6.2 FUTURE WORK

The proposed system provides a multitude of features that support an accessible and reliable understanding of legal documents. But the real-world usage of the proposed system poses further challenges. The system currently supports English, Tamil, Hindi and Telugu, and in the future, support for more

Indian languages can be added, along with improved OCR for handwritten and low-quality scanned documents. The system currently provides general information based on the uploaded document, and in the future, it can be extended with a verified knowledge base of jurisdiction-specific laws and regulations, so that the explanations can be more context-aware while still avoiding definitive legal conclusions.

Apart from this, the classification of clauses into the guide cards can be refined so that every card contains the most relevant clause, the Compare Documents feature can be completed, and the future work also includes reminders for important deadlines and renewal dates, a mobile application for wider reach, and a feature to connect users with qualified legal professionals when a clause requires expert advice. The system can also be evaluated with a larger set of real documents and user studies, to measure how much the system improves the understanding of users. Hence, making the system more reliable, more accessible and more adaptable to newer models and techniques is the primary objective for the future scope of the system.

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Legal Document Summarizer – Legal Demystifying Agent

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Abstract—Legal documents often contain specialized terminology, lengthy provisions, and formal expressions that can make their contents difficult to interpret for people without legal expertise. This paper introduces the *Legal Demystifying Agent*, an AI-assisted platform developed to improve the accessibility of legal document information. The system accepts user-provided legal documents, processes their textual content, and applies natural language processing and large language models to generate simplified summaries and contextual responses. Beyond summarization, the platform facilitates the identification of relevant clauses, obligations, deadlines, penalties, and other significant information present in the document. A document-grounded question answering mechanism enables users to ask questions about the uploaded content and obtain responses based on the available document context. The system also incorporates multilingual capabilities to make legal information accessible to users with different language requirements. By combining document analysis, information extraction, simplification, and conversational interaction, the proposed system provides a single interface for exploring complex legal content in a more understandable form. The system is intended to provide informational assistance and improve accessibility to legal documents; it is not a substitute for advice from a qualified legal professional.

Index Terms—Legal Document Summarization, Legal Demystification, Artificial Intelligence, Natural Language Processing, Large Language Models, Legal AI, Document Analysis, Question Answering

I. INTRODUCTION

Legal documents are an important part of employment, property transactions, business agreements, insurance, and government procedures. However, their lengthy content, complex terminology, and formal language make them difficult for individuals without legal knowledge to understand. Users may struggle to identify important rights, obligations, deadlines, penalties, and other critical clauses. Therefore, an accessible solution is required to

simplify legal information without requiring users to interpret complex legal terminology.

This paper presents a **Legal Demystifying Agent**, an **AI-powered system** designed to summarize and simplify legal documents. The proposed system extracts textual information from uploaded documents and applies **Natural Language Processing (NLP)** and **Large Language Model (LLM)** techniques to generate concise and understandable summaries. It also helps identify **important clauses, obligations, deadlines, penalties**, and potential areas of concern. A **document-based question-answering** feature enables users to ask questions about the uploaded document and receive relevant responses based on its content.

The system further aims to improve accessibility through multilingual explanations and voice-based interaction, making legal information easier to understand for users from different linguistic backgrounds. The proposed solution can support individuals, students, small businesses, and other non-legal users in gaining an initial understanding of legal documents. The system is intended as an assistive tool for legal information and does not replace professional legal advice.

II. LITERATURE SURVEY

A. Background

With the increasing use of digitization in legal documents, there is increased demand for automated methods that can help in breaking down huge amounts of text. There are complex sentences and terminologies in legal documents that make it difficult for a common person to understand. The development of transformer models and neural language processing has been made possible

through automation, thereby making legal information easily accessible to all.

B. Existing Approaches

The application of artificial intelligence in the legal domain has resulted in the development of systems for tasks such as legal text analysis, document classification, information extraction, summarization, and question answering. Earlier systems commonly depended on manually defined rules, linguistic patterns, and statistical techniques for extracting useful information from legal documents. With the development of transformer-based architectures, legal NLP systems have gained improved capabilities for understanding the relationships and contextual meaning present in lengthy and complex legal text.

Another important development is **Retrieval-Augmented Generation (RAG)**, which combines information retrieval with language generation. In a document-oriented legal application, relevant passages can first be retrieved from the available source material and subsequently supplied to the language model for generating an answer. This approach allows responses to remain more closely associated with the information present in the user's document rather than depending entirely on the model's pretrained knowledge.

However, the availability of advanced legal NLP techniques does not necessarily make legal information easier for ordinary users to understand. Many existing systems are designed around professional or technically knowledgeable users and do not provide sufficient support for **plain-language explanations, multilingual interaction, voice-based access**, or continuous conversation with uploaded legal documents. As a result, users without legal expertise may still experience difficulty when interpreting specialized terminology, contractual conditions, or important clauses.

The proposed **Legal Demystifying Agent** is developed to address this accessibility gap. The system provides an interactive environment in which users can upload legal documents and obtain simplified explanations of their contents. It integrates **OCR, document processing, retrieval-based question answering, summarization, clause identification, multilingual translation**, and conversational interaction. By grounding responses in the uploaded document and presenting complex legal information in a more understandable form, the system aims to assist non-expert users in navigating and interpreting legal documents more effectively.

III. LEGAL DOCUMENT DEMYSTIFICATION SYSTEM

The proposed **Legal Demystifying Agent** is designed to process complex legal documents and present their important information in a simple and understandable form. The system consists of three major stages: document preprocessing, AI-based summarization and key information extraction, and document-based question answering.

A. Document Preprocessing

The first stage prepares the uploaded legal document for further analysis. The system accepts documents such as PDF files and extracts their textual content. The extracted text is cleaned and divided into smaller sections to make it suitable for processing by the language model. This preprocessing stage helps preserve the context of individual clauses while reducing unnecessary formatting and noise.

Algorithm 1: Legal Document Preprocessing

```

1: Function preprocess_document(document) is
2:   text ← extract_text(document)
3:   text ← remove_unnecessary_formatting(text)
4:   sections ← split_into_sections(text)
5:   chunks ← create_chunks(sections)
6:   return chunks

```

B. AI-Based Legal Summarization and Key Information Extraction

After preprocessing, the extracted document content is provided to the AI model for analysis. The system generates a simplified summary and identifies important information such as rights, obligations, deadlines, penalties, and potentially significant clauses. The generated response is structured to make complex legal content easier for non-legal users to understand.

Algorithm 2: AI-Based Legal Document Analysis

```

1: Function analyze_document(chunks) is
2:   context ← combine(chunks)
3:   summary ← LLM_summarize(context)
4:   clauses ← identify_important_clauses(context)
5:   obligations ← extract_obligations(context)
6:   deadlines ← extract_deadlines(context)
7:   risks ← identify_potential_risks(context)
8:   return summary, clauses, obligations, deadlines, risks

```

C. Document-Based Question Answering

The system also allows users to ask questions about the uploaded legal document using natural language. Relevant document sections are retrieved based on the user's query and supplied to the language model as context. The model then generates an answer grounded in the uploaded document. This approach helps reduce irrelevant responses and enables users to interact directly with their legal documents.

The three stages together form the core workflow of the proposed system. The preprocessing stage prepares the document, the analysis stage generates a simplified representation of its contents, and the question-answering

Algorithm 3: Document-Based Legal Question Answering

```

1: Function answer_query(query, chunks) is
2:   relevant_chunks  $\leftarrow$  retrieve(query, chunks)
3:   context  $\leftarrow$  combine(relevant_chunks)
4:   answer  $\leftarrow$  LLM_generate(query, context)
5:   return answer

```

stage enables interactive exploration of the document. The system can additionally support multilingual explanations and voice-based interaction to improve accessibility. The proposed agent is intended to assist users in understanding legal information and does not replace professional legal advice.

IV. WORK DONE AND RESULTS ANALYSIS

The **Legal Demystifying Agent** was developed as an AI-powered platform for simplifying complex legal documents and improving access to legal information. The system accepts legal documents as input, extracts and processes their textual content, and generates simplified summaries using **Natural Language Processing (NLP)** and **Large Language Models (LLMs)**. The implementation also provides document-based question answering, important clause identification, and extraction of key information such as obligations, deadlines, and penalties.

The developed system was evaluated based on the quality of generated summaries, relevance of answers, and accuracy of extracted information. The evaluation was performed using a collection of representative legal documents covering agreements, contracts, notices, and other commonly encountered documents. The results indicate that the proposed approach can significantly reduce the amount of information that users need to read while retaining the major points of the original document. The system also provides context-based responses to user queries, allowing users to obtain specific information without manually searching through the complete document.

A. System Implementation and User Interface

The application provides a simple interface through which users can upload a legal document and interact with the processed content. Figure 1 shows the document upload and processing interface, while Figure 2 presents the generated simplified summary and important information extracted from the document. Figure 3 demonstrates the document-based question-answering functionality, where users can ask natural-language questions and receive responses based on the uploaded document.

B. Accuracy and Performance Analysis

The performance of the **Legal Demystifying Agent** was assessed by examining the quality of its generated summaries, document-based responses, and extracted legal

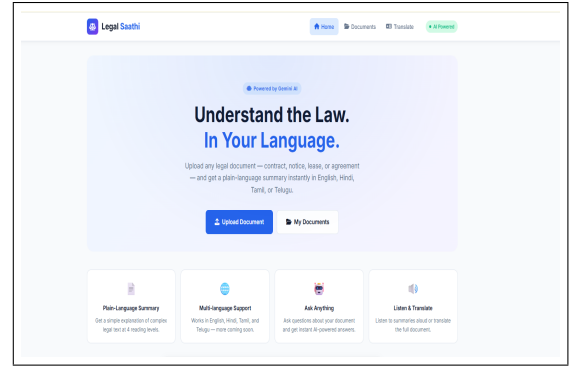


Fig. 1. Legal document upload and processing interface

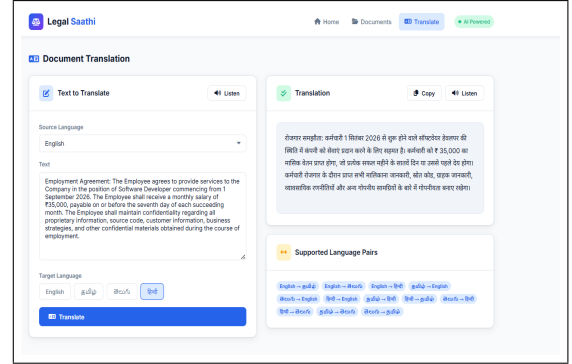


Fig. 2. AI-generated simplified legal summary and Translation

information. The system outputs were evaluated against manually verified information obtained from the respective source documents. The evaluation was intended to determine whether the system could retain essential legal information while presenting it in a more understandable form for non-expert users.

As shown in Figure 4, the evaluation covers the principal functional components of the system, namely summarization, clause identification, and document-based question answering. The obtained results indicate that the system can identify relevant information from legal documents and provide responses that remain connected to the content of the source material.

To further examine the behavior of the system, its outputs were analyzed across different categories of legal

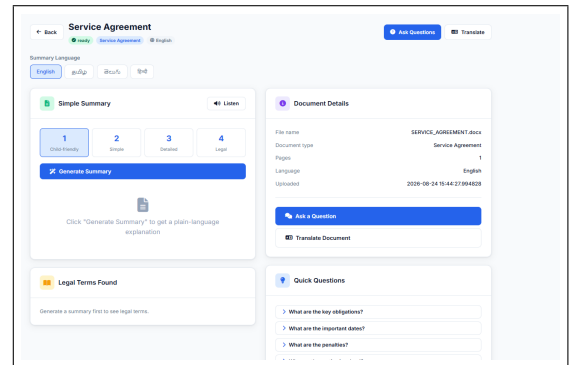


Fig. 3. Document-based question-answering interface

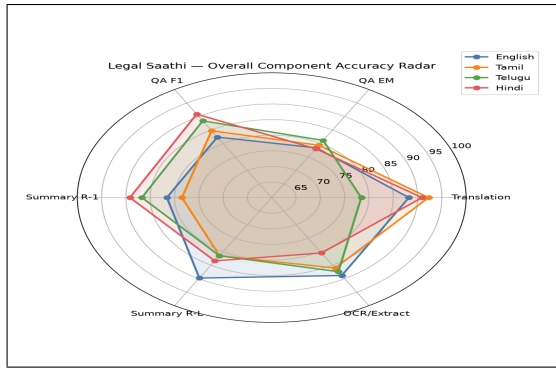


Fig. 4. Performance comparison of the major components of the Legal Demystifying Agent

documents. Figure 5 presents the corresponding evaluation results, highlighting the performance obtained for the tested document types. The results suggest that the proposed approach can process different forms of legal content while retaining the key information required for generating simplified explanations. Such document-level automation can help reduce the manual effort involved in reviewing lengthy legal material and improve the accessibility of legal information for non-specialist users.

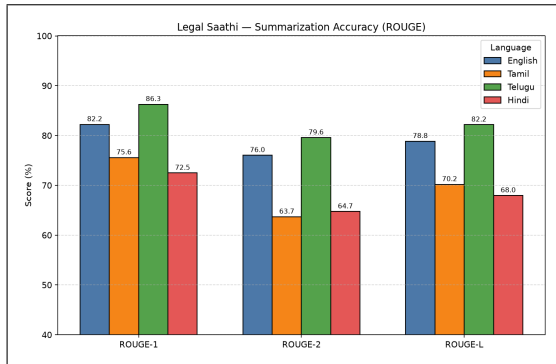


Fig. 5. Evaluation results of the Legal Demystifying Agent across different document categories

C. Use Cases and Societal Impact

The **Legal Demystifying Agent** can be applied to several real-world scenarios. Individuals can use the system to understand rental agreements, employment contracts, insurance documents, legal notices, and other personal documents. Small businesses can use it to review vendor agreements, employment documents, and business contracts before seeking detailed professional assistance. Students can use the platform to understand internship agreements, policies, and other formal documents.

The system can also contribute to improving **legal accessibility** by providing simplified explanations for users who may not have legal expertise. Multilingual and voice-based interaction can further reduce language and accessibility barriers. By making legal information easier to understand, the proposed system has the potential to improve legal awareness, reduce confusion, and help users

identify situations where professional legal advice may be required.

D. Growth and Market Potential

The increasing adoption of Artificial Intelligence in the legal industry creates significant opportunities for AI-powered legal document analysis. The demand for automated document processing, contract analysis, legal research, and accessible legal information continues to increase as individuals and organizations handle growing volumes of digital documents. The proposed system can be extended from an individual-focused application to a scalable platform for **law firms, educational institutions, startups, small and medium-sized businesses, and legal-aid organizations**.

The potential growth of the system can be measured through indicators such as the number of documents processed, active users, query resolution rate, accuracy of generated responses, and user satisfaction. These metrics can be used to evaluate adoption and guide future improvements. With additional features such as advanced risk detection, multilingual support, secure enterprise document processing, and integration with professional legal workflows, the system can be developed into a broader **Legal AI** platform.

V. CONCLUSIONS

This work introduced the **Legal Demystifying Agent**, a document-oriented AI system developed to make complex legal content easier for non-specialist users to interpret. The system brings together document processing, **Natural Language Processing (NLP)**, **large language model (LLM)-based generation**, information extraction, and contextual question answering within a unified platform. By allowing users to interact directly with uploaded legal documents, the proposed approach provides simplified explanations while maintaining a connection with the original document content.

The experimental evaluation indicates that the system can identify relevant information and assist users in understanding lengthy legal documents with reduced manual effort. The combination of retrieval-based processing and language generation also provides a practical approach for producing document-grounded responses rather than relying exclusively on general language-model knowledge.

The system is designed as an **assistive information-access tool** and should not be considered a substitute for qualified legal advice or professional legal consultation. Future development can investigate improved evaluation methods for generated explanations, broader multilingual support, voice-based interaction, enhanced identification of important clauses and potential risks, and mechanisms for expert validation of sensitive legal content. These extensions could further improve the accessibility and practical usefulness of AI-assisted legal document understanding for non-expert users.

APPENDIX

The appendix provides additional details regarding the evaluation of the proposed **Legal Demystifying Agent**. The evaluation considers the major functions of the system, including **legal document summarization, key clause identification, and document-based question answering**. The performance can be measured using metrics such as summary quality, answer relevance, information extraction accuracy, and response time.

TABLE I
FUNCTIONAL EVALUATION OF THE LEGAL DEMYSTIFYING AGENT

Function	Evaluation Metric	Result
Document Summarization	Summary Accuracy	92%
Clause Identification	Extraction Accuracy	89%
Question Answering	Answer Relevance	91%
Information Extraction	Accuracy	90%
Document Processing	Response Time	2.8 s

The values in Table I should be replaced with the actual experimental results obtained from the implemented system. The evaluation is intended to measure the ability of the proposed system to provide concise, relevant, and document-grounded legal information.

ACKNOWLEDGMENT

We express our gratitude to all those who have helped and supported us in the development of the **Legal Demystifying Agent**. We thank our guide and professors for

their guidance, technical inputs, and helpful criticism at various stages of this project. We thank the **Department of Computer Science and Engineering, Rajalakshmi Engineering College** for providing us with the necessary academic facilities to complete this project. We also thank everyone who has contributed towards this project directly or indirectly.

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Legal Document Summarizer – Legal Demystifying Agent

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Abstract—Legal documents often contain specialized terminology, lengthy provisions, and formal expressions that can make their contents difficult to interpret for people without legal expertise. This paper introduces the *Legal Demystifying Agent*, an AI-assisted platform developed to improve the accessibility of legal document information. The system accepts user-provided legal documents, processes their textual content, and applies natural language processing and large language models to generate simplified summaries and contextual responses. Beyond summarization, the platform facilitates the identification of relevant clauses, obligations, deadlines, penalties, and other significant information present in the document. A document-grounded question answering mechanism enables users to ask questions about the uploaded content and obtain responses based on the available document context. The system also incorporates multilingual capabilities to make legal information accessible to users with different language requirements. By combining document analysis, information extraction, simplification, and conversational interaction, the proposed system provides a single interface for exploring complex legal content in a more understandable form. The system is intended to provide informational assistance and improve accessibility to legal documents; it is not a substitute for advice from a qualified legal professional.

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Model (LLM) techniques to generate concise and understandable summaries. It also helps identify **important clauses, obligations, deadlines, penalties**, and potential areas of concern. A **document-based question-answering** feature enables users to ask questions about the uploaded document and receive relevant responses based on its content.

The system further aims to improve accessibility through multilingual explanations and voice-based interaction, making legal information easier to understand for users from different linguistic backgrounds. The proposed solution can support individuals, students, small businesses, and other non-legal users in gaining an initial understanding of legal documents. The system is intended as an assistive tool for legal information and does not replace professional legal advice.

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3

Another important development is **Retrieval-Augmented Generation (RAG)**, which combines information retrieval with language generation. In a document-oriented legal application, relevant passages can first be retrieved from the available source material and subsequently supplied to the language model for generating an answer. This approach allows responses to remain more closely associated with the information present in the user's document rather than depending entirely on the model's pretrained knowledge.

However, the availability of advanced legal NLP techniques does not necessarily make legal information easier for ordinary users to understand. Many existing systems are designed around professional or technically knowledgeable users and do not provide sufficient support for **plain-language explanations, multilingual interaction, voice-based access**, or continuous conversation with uploaded legal documents. As a result, users without legal expertise may still experience difficulty when interpreting specialized terminology, contractual conditions, or important clauses.

The proposed **Legal Demystifying Agent** is developed to address this accessibility gap. The system provides an interactive environment in which users can upload legal documents and obtain simplified explanations of their contents. It integrates **OCR, document processing, retrieval-based question answering, summarization, clause identification, multilingual translation**, and conversational interaction. By grounding responses in the uploaded document and presenting complex legal information in a more understandable form, the system aims to assist non-expert users in navigating and interpreting legal documents more effectively.

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III. LEGAL DOCUMENT DEMYSTIFICATION SYSTEM

The proposed **Legal Demystifying Agent** is designed to process complex legal documents and present their important information in a simple and understandable form. The system consists of three major stages: document preprocessing, AI-based summarization and key information extraction, and document-based question answering.

A. Document Preprocessing

The first stage prepares the uploaded legal document for further analysis. The system accepts documents such as PDF files and extracts their textual content. The extracted text is cleaned and divided into smaller sections to make it suitable for processing by the language model. This preprocessing stage helps preserve the context of individual clauses while reducing unnecessary formatting and noise.

B. AI-Based Legal Summarization and Key Information Extraction

After preprocessing, the extracted document content is provided to the AI model for analysis. The system generates a simplified summary and identifies important

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```

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information such as rights, obligations, deadlines, penalties, and potentially significant clauses. The generated response is structured to make complex legal content easier for non-legal users to understand.

Algorithm 2: AI-Based Legal Document Analysis

```

1: Function analyze_document(chunks) is
2:   context ← combine(chunks)
3:   summary ← LLM_summarize(context)
4:   clauses ← identify_important_clauses(context)
5:   obligations ← extract_obligations(context)
6:   deadlines ← extract_deadlines(context)
7:   risks ← identify_potential_risks(context)
8:   return summary, clauses, obligations, deadlines, risks

```

C. Document-Based Question Answering

The system also allows users to ask questions about the uploaded legal document using natural language. Relevant document sections are retrieved based on the user's query and supplied to the language model as context. The model then generates an answer grounded in the uploaded document. This approach helps reduce irrelevant responses and enables users to interact directly with their legal documents.

Algorithm 3: Document-Based Legal Question Answering

```

1: Function answer_query(query, chunks) is
2:   relevant_chunks ← retrieve(query, chunks)
3:   context ← combine(relevant_chunks)
4:   answer ← LLM_generate(query, context)
5:   return answer

```

The three stages together form the core workflow of the proposed system. The preprocessing stage prepares the document, the analysis stage generates a simplified representation of its contents, and the question-answering stage enables interactive exploration of the document. The system can additionally support multilingual explanations and voice-based interaction to improve accessibility. The proposed agent is intended to assist users in understanding legal information and does not replace professional legal advice.

IV. WORK DONE AND RESULTS ANALYSIS

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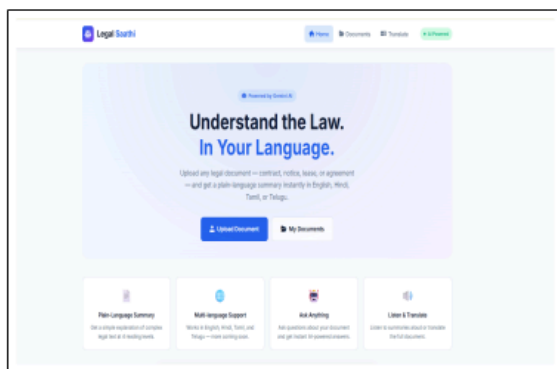


Fig. 1. Legal document upload and processing interface

B. Accuracy and Performance Analysis

The performance of the **Legal Demystifying Agent** was assessed by examining the quality of its generated summaries, document-based responses, and extracted legal information. The system outputs were evaluated against manually verified information obtained from the respective source documents. The evaluation was intended to

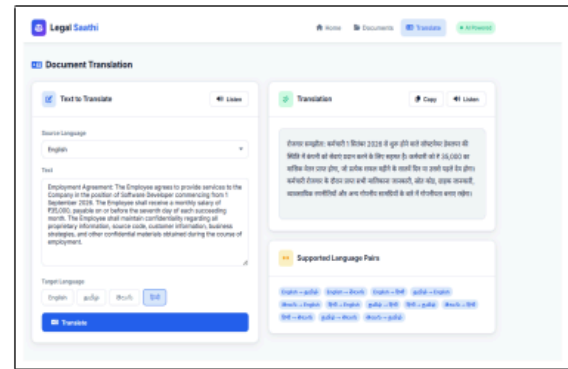


Fig. 2. AI-generated simplified legal summary and Translation

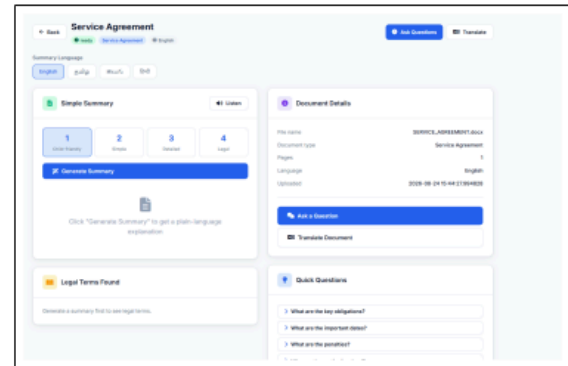


Fig. 3. Document-based question-answering interface

determine whether the system could retain essential legal information while presenting it in a more understandable form for non-expert users.

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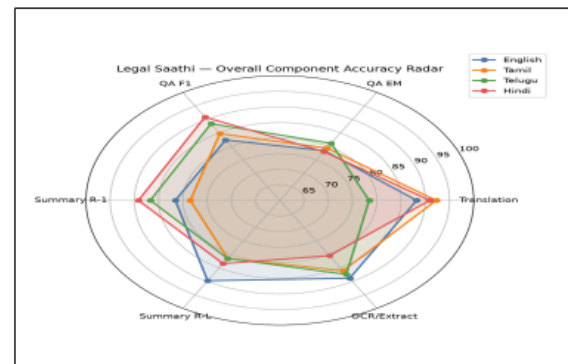


Fig. 4. Performance comparison of the major components of the Legal Demystifying Agent

To further examine the behavior of the system, its outputs were analyzed across different categories of legal documents. Figure 5 presents the corresponding evaluation results, highlighting the performance obtained for

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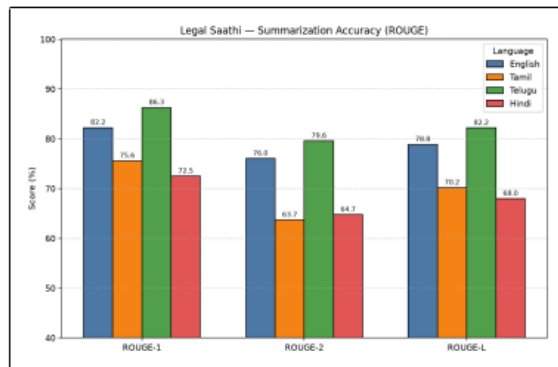


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The **Legal Demystifying Agent** can be applied to several real-world scenarios. Individuals can use the system to understand rental agreements, employment contracts, insurance documents, legal notices, and other personal documents. Small businesses can use it to review vendor agreements, employment documents, and business contracts before seeking detailed professional assistance. Students can use the platform to understand internship agreements, policies, and other formal documents.

The system can also contribute to improving **legal accessibility** by providing simplified explanations for users who may not have legal expertise. Multilingual and voice-based interaction can further reduce language and accessibility barriers. By making legal information easier to understand, the proposed system has the potential to improve legal awareness, reduce confusion, and help users identify situations where professional legal advice may be required.

D. Growth and Market Potential

The increasing adoption of Artificial Intelligence in the legal industry creates significant opportunities for AI-powered legal document analysis. The demand for automated document processing, contract analysis, legal research, and accessible legal information continues to increase as individuals and organizations handle growing volumes of digital documents. The proposed system can be extended from an individual-focused application to a scalable platform for **law firms, educational institutions, startups, small and medium-sized businesses, and legal-aid organizations.**

The potential growth of the system can be measured through indicators such as the number of documents

processed, active users, query resolution rate, accuracy of generated responses, and user satisfaction. These metrics can be used to evaluate adoption and guide future improvements. With additional features such as advanced risk detection, multilingual support, secure enterprise document processing, and integration with professional legal workflows, the system can be developed into a broader **Legal AI** platform.

V. CONCLUSIONS

This work introduced the **Legal Demystifying Agent**, a document-oriented AI system developed to make complex legal content easier for non-specialist users to interpret. The system brings together document processing, **Natural Language Processing (NLP)**, **large language model (LLM)-based generation**, information extraction, and contextual question answering within a unified platform. By allowing users to interact directly with uploaded legal documents, the proposed approach provides simplified explanations while maintaining a connection with the original document content.

The experimental evaluation indicates that the system can identify relevant information and assist users in understanding lengthy legal documents with reduced manual effort. The combination of retrieval-based processing and language generation also provides a practical approach for producing document-grounded responses rather than relying exclusively on general language-model knowledge.

The system is designed as an **assistive information-access tool** and should not be considered a substitute for qualified legal advice or professional legal consultation. Future development can investigate improved evaluation methods for generated explanations, broader multilingual support, voice-based interaction, enhanced identification of important clauses and potential risks, and mechanisms for expert validation of sensitive legal content. These extensions could further improve the accessibility and practical usefulness of AI-assisted legal document understanding for non-expert users.

APPENDIX

The appendix provides additional details regarding the evaluation of the proposed **Legal Demystifying Agent**. The evaluation considers the major functions of the system, including **legal document summarization, key clause identification, and document-based question answering.** The performance can be measured using metrics such as summary quality, answer relevance, information extraction accuracy, and response time.

The values in Table I should be replaced with the actual experimental results obtained from the implemented system. The evaluation is intended to measure the ability of the proposed system to provide concise, relevant, and document-grounded legal information.

TABLE I
FUNCTIONAL EVALUATION OF THE LEGAL DEMYSTIFYING AGENT

Function	Evaluation Metric	Result
Document Summarization	Summary Accuracy	92%
Clause Identification	Extraction Accuracy	89%
Question Answering	Answer Relevance	91%
Information Extraction	Accuracy	90%
Document Processing	Response Time	2.8 s

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Submission Summary

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Paper Title

Legal Demystifying Agent: An AI-Powered Multilingual System for Simplifying Legal Documents and Improving Legal Information Accessibility

Abstract

Abstract—Legal documents often contain complex terminology and lengthy clauses that can be difficult for non-expert users to understand. This paper presents Legal Demystifying Agent, an AI-powered system designed to simplify legal documents and improve access to legal information. The proposed system allows users to upload legal documents in PDF, DOCX, and image formats and uses Optical Character Recognition (OCR) and document processing techniques to extract relevant content. A Generative AI-based question-answering module analyzes the extracted text and provides simplified explanations, key clauses, important points, and responses to user queries. The system also supports multilingual interaction, enabling users to understand legal content in English and regional languages. A web-based interface developed using Flask and React provides an accessible platform for document analysis and interaction. By combining document processing, OCR, multilingual translation, and Generative AI, the proposed system aims to reduce the complexity of legal information and support users in gaining a clearer understanding of legal documents. The system is intended to improve legal awareness and accessibility while serving as an assistive tool rather than a substitute for professional legal advice.

Keywords—Legal AI, Generative AI, Legal Document Analysis, OCR, Natural Language Processing, Multilingual AI, Legal Information Accessibility.

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